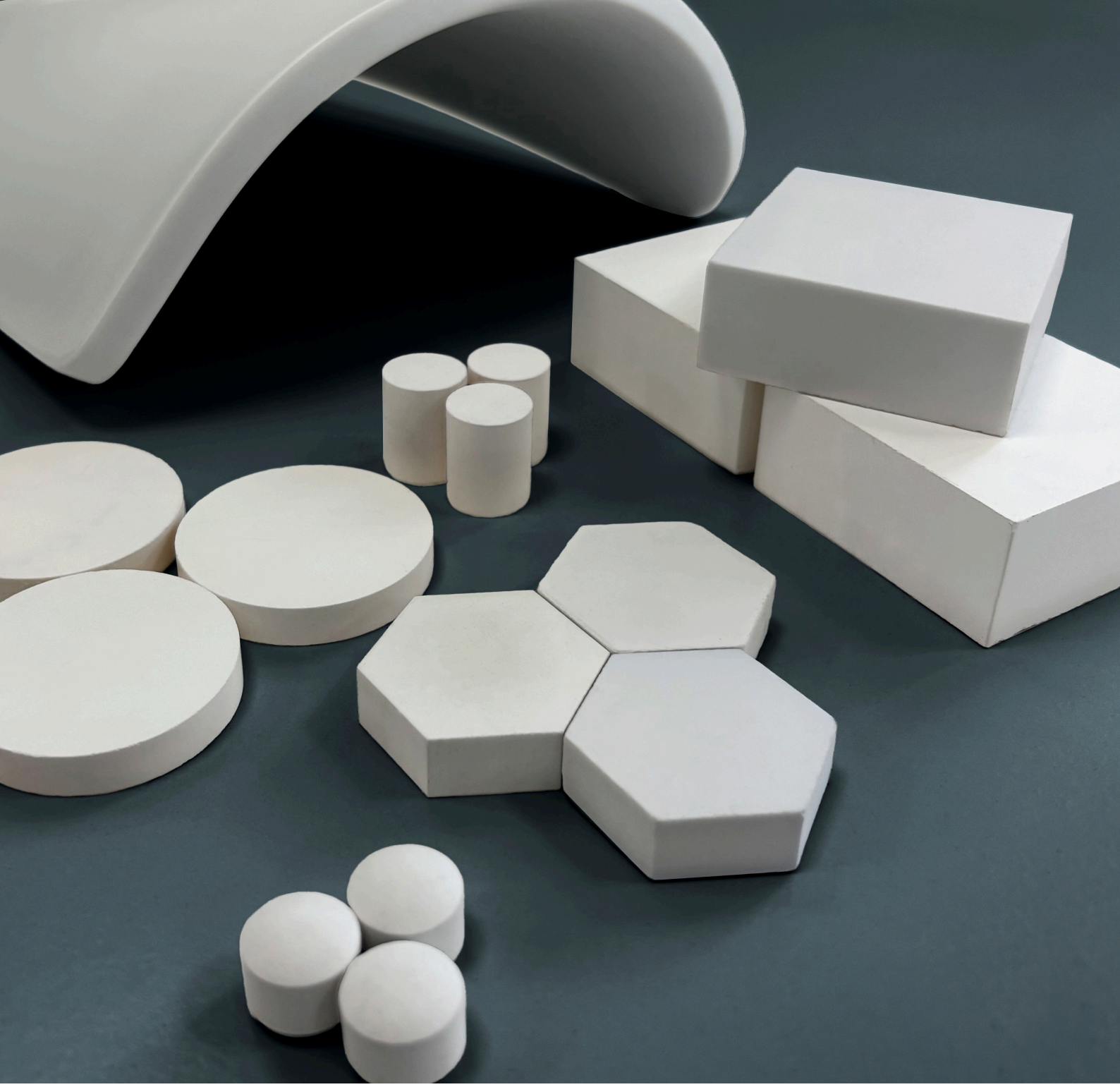


ADVANCED BALLISTIC CERAMICS

High-performance ceramic materials
designed for ballistic protection systems.

Engineered and manufactured in France.

BERNARDAUD
SOLUTIONS
INDUSTRIELLES

BERNARDAUD

Tradition and innovation since 1863

BUILT FOR DEMANDING APPLICATIONS

Founded in Limoges in 1863, Bernardaud is a French, family-owned ceramic manufacturer combining long-standing material expertise with modern industrial capabilities.

Our facilities bring together ceramic formulation, prototyping, forming, sintering, machining, finishing and characterization. This integrated approach allows us to support customers from initial design through qualification and serial production.

KEY FIGURES

160 years of history

750 employees

4 ceramic production plants
located in the Limoges region, in France

60 000 m² dedicated to production

ISO 9001 – ISO 14001



Oradour-sur-Glane



Limoges - City center



Aixe-sur-Vienne



Limoges - North industrial area

An industrial site, unique in France

FROM PROTOTYPING TO SERIAL PRODUCTION

Our capabilities include:

- Axial and semi-isostatic pressing
- Ceramic injection moulding
- Casting and die casting
- Waterjet cutting and robotic finishing
- Debinding and high-temperature sintering
- Surface treatment and texturing
- Material, mechanical and functional characterization

DEVELOPMENT AND VALIDATION CAPABILITIES

- Material formulation and selection
- Design for manufacturing
- Rapid prototyping
- Material and microstructure analysis
- Density and hardness measurement
- Mechanical and ultrasonic testing
- Surface inspection
- Comparative ballistic testing
- Industrialization and serial production

BALLISTIC PROTECTION
Structure

LIGHTWEIGHT PROTECTION AGAINST HIGH-LEVEL THREATS

Ballistic ceramics are used to provide protection against high-velocity projectiles while limiting the weight of the complete armour system. Their extreme hardness damages and fractures the projectile on impact. The ceramic also helps in a controlled manner, to disperse the impact energy over a wider area. The ceramic is combined with a backing material (typically polyethylene, aramid or composite) to capture fragments and absorb the remaining energy.

HOW A CERAMIC ARMOUR SYSTEM WORKS



Hard ceramic plate + Backing

Integrated protection plate (strike face)

Hard body armour

1. Hard ceramic plate
Breaks the projectile and disperses its energy. Material: Alumina, Zirconia

2. Backing
Captures fragments and absorbs the residual impact. Material: PE, Kevlar

3. Complete body armour system
Provides the required protection and ergonomic levels while optimizing weight, thickness and cost.

Final applications

Soldier
Law Enforcement
Special unit Forces

Hard body armour plates
Ballistic shields



Armoured vehicles
Command & logistic trucks
Turrets

Hard side armour panels
Mine floor protection
Roof protection
Gunner protection



Armoured ships
Coast guard boats

Pilot cabin protection
Crew cabin protection
Turret protection
Gunner protection



Aircrafts (fixed wing)
Helicopters (rotary wing)

Cockpit floor
Side protection panels



Radar
Mobile post command

Casing
Side panels



BALLISTIC PRODUCTS
Ceramic components

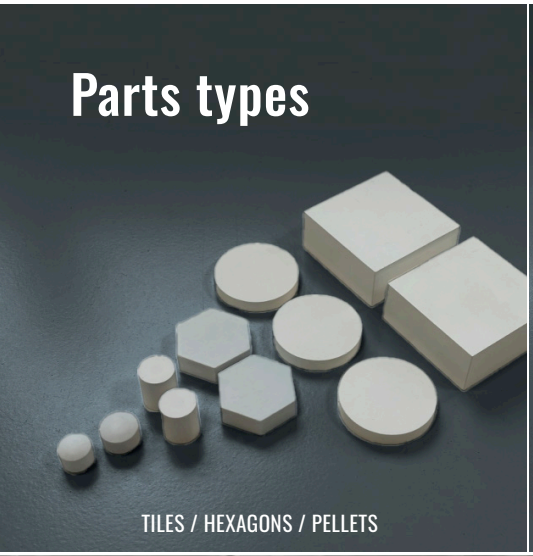
WHERE BERNARDAUD FITS

Bernardaud develops and manufactures advanced ceramic components that form the strike face of the armour system. We offer standard shapes and material formulations, or fully customized developments tailored to armed forces requirements.

MATERIALS

HIGH PURITY ALUMINIA
ZTA
ZIRCONIA

Parts types



TILES / HEXAGONS / PELLETS



MONOLITHICS PLATES 250 × 300 – Thickness 3.5 to 12mm



MULTI CURVATURE PARTS

Range of materials & performances

MATERIAL	PURITY (%)	DENSITY ($g \cdot cm^{-3}$)	HARDNESS (Rockwell45N)	TENACITY K_IC (MPA·M^0.5)	YOUNG MODULUS (GPa)	ULTRASONIC SPEED (m/s)
Al_2O_3 96	96 %	3,75	80	4-5	300	
Al_2O_3 98	98 %	3,85	84	4-5	360	10400
Al_2O_3 99,7	99,7 %	3,91	85	4-5	370	10400
ZTA (Al_2O_3 + ZrO_2)	90/10	4,15	92	5-6,5	380	10600

Bernardaud components are suitable for **VPAM, NIJ III + and NIJ IV** protection standards.

	30.06 AP M (perforante) - 7,62x54 R API B32 (perforante)	**NIJ 4**
	7,62x39 API BZ (perforante)	**NIJ 3++**
	7,62x54 R LPS (semi-perforante)	**NIJ 3++**
	7,62x39 API BZ (perforante)	**NIJ 3++**
	7,62x39 MSC (semi-perforante) - 5.56x45 M193	**NIJ 3+**
	7,62x51 - NATO FMJ	**NIJ 3**
	357 FMJ/FN - 357 Magnum FMs/CB	**NIJ 3A**

BALLISTIC PRODUCTS
Value proposition

HIGH-PERFORMANCE BALLISTIC CERAMICS

We manufacture high-performance ceramic components for integration into armour systems targeting NIJ Level III and Level IV protection.

CUSTOM ENGINEERING AND INNOVATION

We develop bespoke materials, dimensions, thicknesses, curvatures and complex geometries to meet new protection requirements.

FLEXIBILITY AND RESPONSIVENESS

Our integrated teams rapidly assess requirements, produce prototypes, refine designs and manufacture qualification batches.

SCALABLE MANUFACTURING

Our large-scale European production capacity supports both specialized batches and high-volume requirements, with short and reliable lead times.

MADE IN FRANCE

Development and production are carried out at our facilities in France, providing a secure European supply chain and ITAR-free products. Our company is family owned and managed.

QUALITY AND TRACEABILITY

From raw-material preparation to final inspection, manufacturing, traceability and characterization are controlled within our ISO 9001 and ISO 14001 certified management systems.





«Responsiveness, continuous innovation, and industrial independence are now essential to meeting today's defense challenges. From the heart of France, Bernardaud provides a robust industrial capability, combining manufacturing excellence with uncompromising quality to support the evolving needs of defense and security forces.»

Michel Bernardaud
Chairman and CEO



Standard products when speed matters.
Bespoke development when the application demands it.

LET'S DEVELOP YOUR NEXT PROTECTION SOLUTION!

BERNARDAUD
SOLUTIONS
INDUSTRIELLES

Contact:
Email : contact@bernardaud.com
Tel. : +33 (0)5 55 10 55 50

BERNARDAUD
19 rue Pierre Bernardaud, 87100 Limoges, France

ISO 9001
ISO 14001
SOCOTEC
CERTIFIÉ